PROJECT CASE STUDY

# Explainable Brain Stimulation-Response AI

3D fMRI Transfer Learning + Multimodal Decision Support

| Category | Status | Year |
| --- | --- | --- |
| AI Research | Ongoing research | 2026 |

A leakage-aware multimodal framework combining pretrained 3D fMRI representations, stimulation/electrode metadata, uncertainty, visual interpretation, and grounded report generation.

| 159 | 26 | 42.05% |
| --- | --- | --- |
| epilepsy runs | LOSO subjects | macro-F1 |

## Context and problem

The epilepsy dataset contains only 159 CNN-compatible runs from 26 subjects, making random-initialization 3D CNN training unreliable and requiring strict subject-wise evaluation.

## Strategy and implementation

- Pretrained a residual 3D autoencoder on 6,562 BrainPedia statistical maps.
- Continued self-supervised pretraining on 17,797 HCP task-fMRI maps from 787 subjects.
- Fine-tuned a 64-dimensional encoder on Low/Moderate/High pseudo-labels using LOSO.
- Integrated fMRI evidence with stimulation/electrode metadata, uncertainty, explainability, VLM interpretation, and grounded LLM reporting.

## My role

Research pipeline design, preprocessing, transfer learning, subject-wise validation, fusion, uncertainty, explainability, and reporting architecture.

## Technology stack

PyTorch | 3D CNN | fMRI | NIfTI | BrainPedia | HCP | LOSO | VLM | LLM

## Results and evidence

- Best documented subject-wise result: approximately 41.82% balanced accuracy and 42.05% macro-F1.
- Established a 6,562 -> 17,797 -> 159 transfer-learning story.
- Built an end-to-end research decision-support framework rather than claiming clinical validation.

## Key takeaways

- Subject-wise evaluation is mandatory when multiple runs come from the same person.
- Pseudo-label uncertainty can be more important than simply making the CNN larger.

**Research note:** This project is presented with conservative claims. Proposed architecture elements and prototype evidence are separated from results that still require broader comparison or validation.

Portfolio: github.com/awai1s | awaisraza5424@gmail.com